

HDOC DAY - 10

Question 1:

Write a program to input time in seconds and convert it into days, hours, minutes, and seconds format.

Algorithm

1. Start.
2. Input total seconds.
3. Calculate days = seconds / 86400.
4. Calculate remaining seconds.
5. Calculate hours = remaining seconds / 3600.
6. Calculate minutes = remaining seconds / 60.
7. Remaining value is seconds.
8. Display the result.
9. Stop.

Pseudocode:

START

INPUT totalSeconds

days = totalSeconds / 86400

totalSeconds = totalSeconds % 86400

hours = totalSeconds / 3600

totalSeconds = totalSeconds % 3600

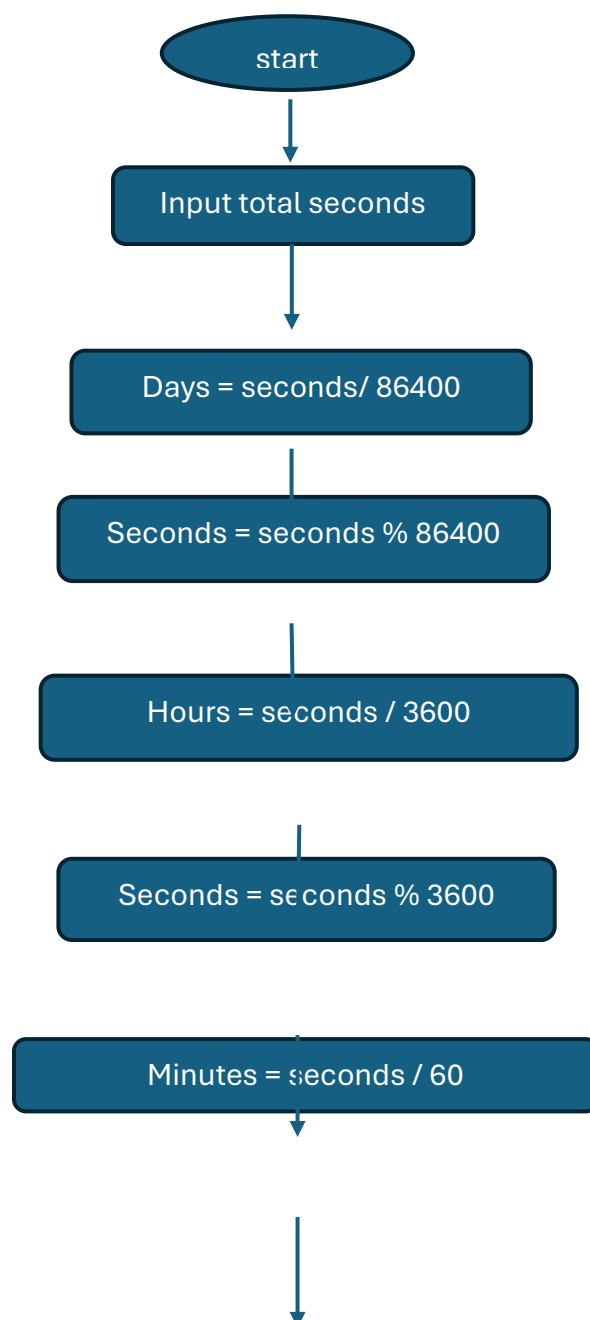
$\text{minutes} = \text{totalSeconds} / 60$

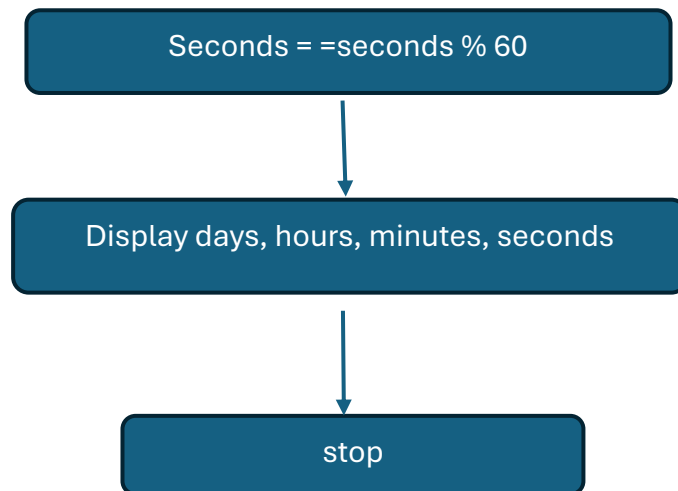
$\text{seconds} = \text{totalSeconds} \% 60$

PRINT days, hours, minutes, seconds

STOP

Flowchart:





C program:

```
#include <stdio.h>
```

```
int main() {
```

```
    long int totalSeconds, days, hours, minutes, seconds;
```

```
    printf("Enter time in seconds: ");
```

```
    scanf("%ld", &totalSeconds);
```

```
    days = totalSeconds / 86400;
```

```
    totalSeconds = totalSeconds % 86400;
```

```
    hours = totalSeconds / 3600;
```

```
    totalSeconds = totalSeconds % 3600;
```

```
    minutes = totalSeconds / 60;
```

```
    seconds = totalSeconds % 60;
```

```
printf("\nTime = %ld days, %ld hours, %ld minutes, %ld seconds",  
      days, hours, minutes, seconds);
```

```
return 0;
```

```
}
```

```
#include <stdio.h>  
  
int main() {  
    long int totalSeconds, days, hours, minutes, seconds;  
  
    printf("Enter time in seconds: ");  
    scanf("%ld", &totalSeconds);  
  
    days = totalSeconds / 86400;  
    totalSeconds = totalSeconds % 86400;  
  
    hours = totalSeconds / 3600;  
    totalSeconds = totalSeconds % 3600;  
  
    minutes = totalSeconds / 60;  
    seconds = totalSeconds % 60;  
  
    printf("\nTime = %ld days, %ld hours, %ld minutes, %ld seconds",  
          days, hours, minutes, seconds);  
  
    return 0;  
}
```

Execution and compilation:

```
(kali㉿kali)-[~]  
$ vi h1.c  
  
(kali㉿kali)-[~]  
$ gcc h1.c -o h1  
  
(kali㉿kali)-[~]  
$ ./h1  
Enter time in seconds: 60  
Time = 0 days, 0 hours, 1 minutes, 0 seconds
```

Test case:

Input(seconds)	Expected output
60	0 days, 0 hours, 1 minute, 0 seconds
3600	0 days, 1 hour, 0 minutes, 0 seconds

90061	1 day, 1 hour, 1 minute, 1 second
172800	2 days, 0 hours, 0 minutes, 0 seconds

```

(kali㉿kali)-[~]
$ ./h1
Enter time in seconds: 60

Time = 0 days, 0 hours, 1 minutes, 0 seconds

(kali㉿kali)-[~]
$ ./h1
Enter time in seconds: 3600

Time = 0 days, 1 hours, 0 minutes, 0 seconds

(kali㉿kali)-[~]
$ ./h1
Enter time in seconds: 90061

Time = 1 days, 1 hours, 1 minutes, 1 seconds

(kali㉿kali)-[~]
$ ./h1
Enter time in seconds: 172800

Time = 2 days, 0 hours, 0 minutes, 0 seconds

```

Question 2:

Write a program to input a year and check whether it is a leap year or not

Algorithm

1. Start.
2. Input a year.
3. Check if the year is divisible by 400.
4. If yes, it is a leap year.
5. Otherwise, check whether it is divisible by 4 and not divisible by 100.
6. If yes, it is a leap year; otherwise, it is not.
7. Display the result.
8. Stop.

Pseudocode:

START

INPUT year

IF year % 400 == 0

 PRINT "Leap Year"

ELSE IF year % 4 == 0 AND year % 100 != 0

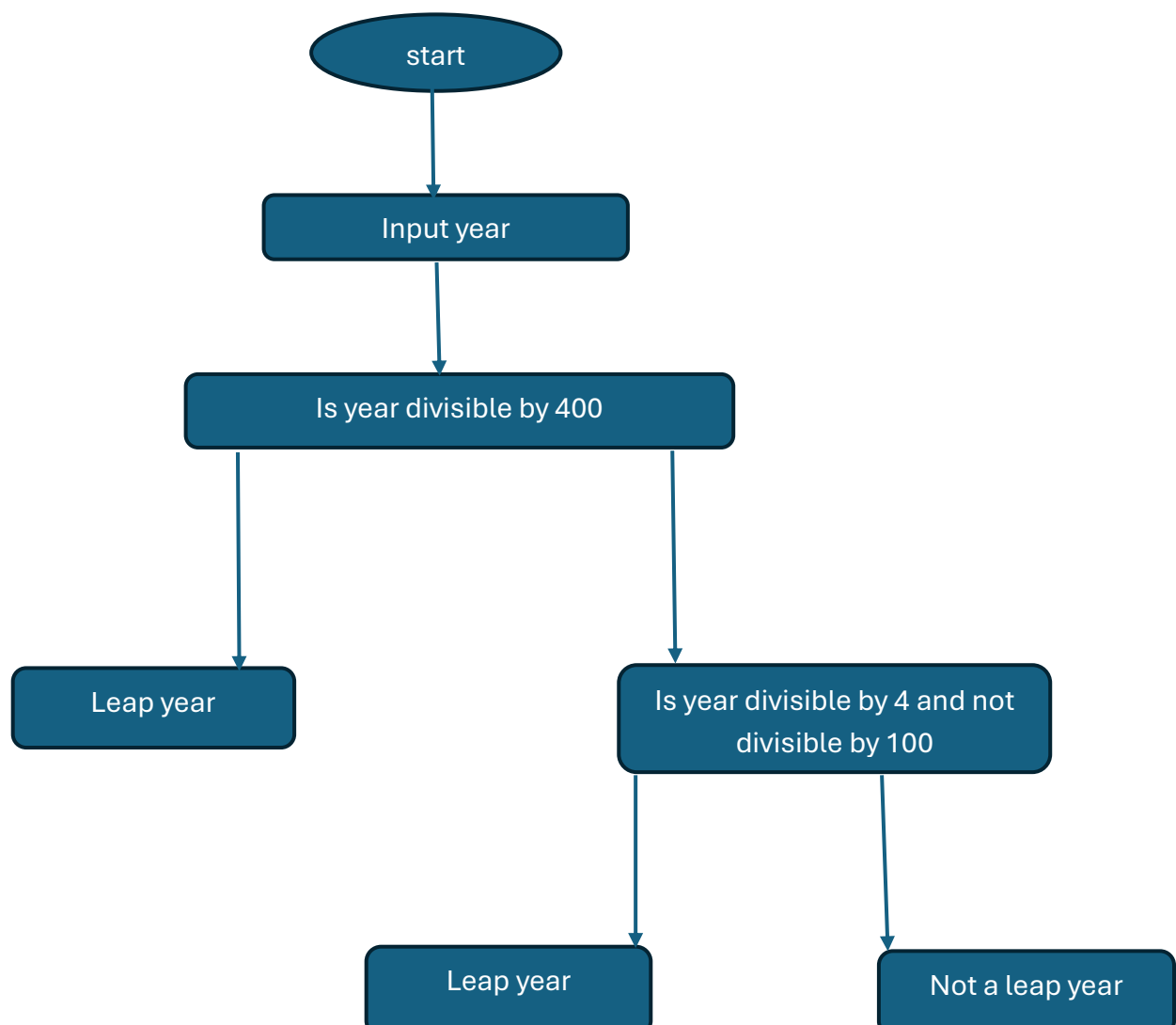
 PRINT "Leap Year"

ELSE

 PRINT "Not a Leap Year"

STOP

Flowchart:



C program:

```
#include <stdio.h>
```

```
int main() {
```

```
    int year;
```

```
    printf("Enter a year: ");
```

```
    scanf("%d", &year);
```

```
    if (year % 400 == 0) {
```

```
        printf("%d is a leap year.", year);
```

```
    }
```

```
    else if (year % 4 == 0 && year % 100 != 0) {
```

```
        printf("%d is a leap year.", year);
```

```
    }
```

```

else {

    printf("%d is not a leap year.", year);

}

return 0;

}

```

```

#include <stdio.h>

int main() {
    int year;

    printf("Enter a year: ");
    scanf("%d", &year);

    if (year % 400 == 0) {
        printf("%d is a leap year.", year);
    }
    else if (year % 4 == 0 && year % 100 != 0) {
        printf("%d is a leap year.", year);
    }
    else {
        printf("%d is not a leap year.", year);
    }

    return 0;
}

```

Execution and compilation:

```

(kali㉿kali)-[~]
$ vi h2.c

(kali㉿kali)-[~]
$ gcc h2.c -o h2

(kali㉿kali)-[~]
$ ./h2
Enter a year: 2024
2024 is a leap year.

```

Test case:

Input(s)	Expected result	Reason
2024	Leap year	Divisible by 4
2022	Not a leap year	Not divisible by 4
1994	Not a leap year	Not divisible by 4
2002	Not a leap year	Not divisible by 4

